

Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

1 Generating synthetic financial time series that preserve the statistical prop-
2 erties of real market data is essential for stress testing, risk model validation,
3 and scenario design, yet existing approaches—from parametric models to
4 deep generative networks—struggle to simultaneously reproduce the heavy-
5 tailed distributions, negligible linear autocorrelation, and persistent volatility
6 clustering that characterize empirical equity data. We propose a hybrid hid-
7 den Markov framework that discretizes continuous excess growth rates into
8 Laplace quantile-defined market states and augments the regime-switching
9 process with a Poisson-driven jump-duration mechanism to enforce realistic
10 tail-state dwell times. Parameters are estimated by direct transition count-
11 ing, avoiding the Baum-Welch EM algorithm entirely. We evaluate synthetic
12 data quality using three complementary metrics: Kolmogorov-Smirnov and
13 Anderson-Darling goodness-of-fit pass rates for distributional fidelity, and
14 ACF mean absolute error for temporal structure preservation. Applied to ten
15 years of SPY data across 1,000 simulated paths, the framework achieves KS
16 and AD pass rates exceeding 97% and 90% in-sample and 95% out-of-sample
17 (full calendar year 2025), reproducing the ARCH effect that standard regime-
18 switching models miss. GARCH(1,1), by contrast, achieves only 4.6% and
19 1.1% in-sample pass rates despite its well-known volatility clustering prop-
20 erties. A Single-Index Model extension propagates the SPY factor path to
21 a 424-asset universe, enabling scalable correlated synthetic path generation
22 while preserving cross-sectional correlation structure.

Keywords: Synthetic Data Quality, Time Series Generation, Hidden
Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching,

23 Introduction

24 The generation of synthetic financial time series that faithfully preserve
25 the statistical properties of real market data is a central challenge in quanti-
26 tative finance and data quality research. High-fidelity synthetic data are es-
27 sential for stress testing risk models against plausible but unobserved market
28 scenarios, validating portfolio optimization algorithms, and augmenting lim-
29 ited training sets for machine learning pipelines [1, 2]. Yet the quality require-
30 ments for financial synthetic data are unusually demanding: empirical equity
31 excess growth rates exhibit three well-documented statistical regularities—
32 heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in
33 raw returns, and persistent volatility clustering—that any generative frame-
34 work must simultaneously reproduce to be considered statistically faithful
35 [3–5]. These domain-specific quality criteria, often called *stylized facts*, serve
36 as the benchmark against which the fidelity of synthetic financial data must
37 be evaluated.

38 Existing generative approaches address these quality requirements from
39 different angles but each falls short on at least one dimension. GARCH-
40 family models [6, 7] capture conditional variance dynamics but do not explic-
41 itly represent discrete market regimes or sudden jumps in volatility. Stochastic-
42 volatility and jump-diffusion models [8, 9] enrich tail behavior but lack a
43 mechanism for volatility persistence after jumps. Deep generative models,
44 including recurrent neural networks [10, 11] and generative adversarial net-
45 works [12–14], can learn complex distributions from data but struggle to re-
46 produce temporal dependence structures—particularly volatility clustering—
47 without explicitly encoding the relevant inductive biases [5]. Hidden Markov
48 models (HMMs) and related regime-switching frameworks offer a probabilis-
49 tic structure for non-stationary market behavior through latent state dy-
50 namics [15], yet standard HMM specifications fail to generate persistent
51 high-volatility regimes, leading to overly rapid reversion following extreme
52 events and poor temporal quality in the synthetic output [16, 17]. Bridging
53 this gap requires a generative framework that preserves both distributional
54 fidelity and temporal structure while remaining computationally scalable.

55 We address this challenge by developing a hybrid HMM framework that
56 discretizes continuous excess growth rates into quantile-based regimes and

57 augments the resulting Markov process with a jump-duration mechanism
 58 that enforces volatility persistence, linking smooth diffusion to discontinuous
 59 shocks identified by jump-detection methods [18]. Using a 424-asset dataset
 60 with SPY as a case study, we estimate transition dynamics empirically—
 61 avoiding the EM algorithm entirely—and validate the model through in-
 62 sample and out-of-sample tests on distributional fit and volatility clustering.
 63 We further extend the univariate framework to a correlated multi-asset set-
 64 ting via a Single-Index Model (SIM) factor structure [19], enabling synthetic
 65 path generation across all 424 assets in a single generative pass.

66 This paper makes four contributions to synthetic financial data genera-
 67 tion and quality assessment. First, we propose augmenting a discrete-state
 68 regime-switching model with a two-parameter Poisson jump-duration mech-
 69 anism (ϵ, λ) that forces the model to dwell in high-volatility tail states for
 70 empirically realistic durations, providing a structural inductive bias that pre-
 71 serves the temporal quality (volatility clustering) of the synthetic output—
 72 addressing the key limitation identified in prior evaluations of both HMM-
 73 based [16] and GAN-based [13] financial data generators. Second, by parti-
 74 tioning excess growth rates into quantile-defined states via a Laplace CDF,
 75 we enable direct frequentist counting of regime transitions, eliminating the
 76 computational cost and initialization sensitivity of the Baum-Welch EM al-
 77 gorithm and making the framework scalable to a 424-asset synthetic data
 78 pipeline. Third, we conduct a rigorous quality evaluation using three com-
 79plementary metrics—Kolmogorov-Smirnov and Anderson-Darling pass rates
 80 for distributional fidelity, and ACF mean absolute error for temporal struc-
 81 ture preservation—all reported with standard errors. Using ten years of SPY
 82 data (2014–2024) for training and 249 trading days (full calendar year 2025)
 83 for out-of-sample testing, we demonstrate KS and AD pass rates of 97.1%
 84 (SE 0.5) and 90.4% (SE 0.9) in-sample and 95.0% (SE 0.7) and 95.4% (SE
 85 0.7) out-of-sample across 1,000 simulated paths, with volatility clustering
 86 preserved in both windows. Finally, we extend the univariate framework
 87 to the full 424-asset universe through a Single-Index Model factor projec-
 88 tion, enabling scalable correlated synthetic path generation while preserving
 89 cross-sectional correlation structure.

90 We find that a state resolution of $N = 100$ provides a balance between
 91 distributional fidelity and statistical robustness, and that the jump-duration
 92 extension is necessary for reproducing persistent high-volatility regimes that
 93 standard regime-switching models miss.

94 Related Work

95 *Classical stylized facts and the failure of Gaussian diffusion*

96 The quantitative study of financial market dynamics has been shaped by
97 a tension between tractable theoretical frameworks and the empirical com-
98 plexity of observed data. The Black-Scholes-Merton option pricing model [20]
99 and the geometric Brownian motion (GBM) paradigm embedded within it
100 assume that log-returns are independently and identically distributed Gaus-
101 sian random variables. Under this assumption, asset prices follow a diffusion
102 with constant drift and volatility, price paths are almost surely continuous,
103 and the tails of the return distribution decay exponentially. These proper-
104 ties have well-known practical advantages: GBM admits closed-form option
105 prices, portfolio variances aggregate linearly, and model parameters can be
106 estimated with minimal data.

107 However, the Gaussian diffusion paradigm conflicts with a broad set of
108 empirical regularities that were documented well before the widespread adop-
109 tion of the model. Mandelbrot [3] demonstrated that the distributions of
110 speculative price changes exhibit excess kurtosis—the probability of large
111 deviations from the mean is far higher than Gaussian predictions, a phe-
112 nomenon now described as *fat tails* or *leptokurtosis*. Fama [21] showed that,
113 consistent with the efficient markets hypothesis, raw return series display
114 negligible linear autocorrelation; daily or weekly returns are essentially un-
115 predictable from their own past values. Yet Schwert [22] and others doc-
116 umented that the *magnitude* of returns—measured by absolute or squared
117 returns—exhibits substantial positive autocorrelation that decays slowly over
118 time. This temporal clustering of variance, where large price changes tend
119 to be followed by large changes regardless of sign, is referred to as *volatility*
120 *clustering* and is among the most persistently observed features of financial
121 time series. Cont [4] provides a systematic review of these and related styl-
122 ized facts, confirming their presence across asset classes, geographic markets,
123 and sampling frequencies. Collectively, leptokurtosis, the absence of return
124 predictability, and volatility clustering define the benchmark against which
125 any proposed model of market dynamics must be evaluated.

126 *Conditional variance models: ARCH and GARCH families*

127 The most influential class of models designed to address volatility clus-
128 tering within the Gaussian framework is the autoregressive conditional het-
129 eroscedasticity (ARCH) family introduced by Engle [6]. The ARCH model

conditions the variance of the current observation on the squared residuals of past observations, allowing variance to cluster without introducing autocorrelation in the level series. Bollerslev [7] generalized this to the GARCH(p,q) framework, in which the conditional variance depends on both past squared residuals and past conditional variances; the GARCH(1,1) specification became the canonical one-factor volatility model and remains widely used in practice. GARCH models are parsimonious, analytically tractable, and can be estimated by maximum likelihood under distributional assumptions on the innovation.

Despite their success in modeling conditional variance dynamics, standard GARCH models have several well-documented limitations in the context of stylized-facts reproduction. First, while GARCH innovations can be made leptokurtic by specifying Student-t or generalized error distributions, the fat-tail behavior arises through the parametric distributional assumption rather than through the structural dynamics of the model. Second, GARCH models do not explicitly represent discrete market regimes or the possibility of sudden jumps in the volatility process; the variance responds smoothly and continuously to past shocks. Third, regime persistence—the tendency of markets to remain in high-volatility states for sustained periods—is captured only indirectly through the persistence parameter of GARCH, which governs exponential decay of the volatility response. When persistence is near unity, the GARCH model is near-integrated and exhibits long-memory-like behavior, but this is a parametric approximation rather than a structural representation of distinct market regimes [23]. Stochastic volatility models [8, 24] address some of these shortcomings by treating variance as a latent diffusion process, but they require computationally intensive estimation via particle filters or Markov chain Monte Carlo methods, and their parameters do not directly correspond to observable market states.

Jump-diffusion models

An alternative approach to modeling fat tails and discontinuous price behavior is the explicit introduction of Poisson-driven jumps into the asset price process. Merton [9] extended the Black-Scholes framework by adding a compound Poisson jump component to the GBM, producing a model in which prices follow a diffusion between events and undergo discrete jumps at random times governed by a Poisson arrival process. This construction directly addresses leptokurtosis: the marginal distribution of returns is a Gaussian mixture weighted by the Poisson probability of observing zero, one, or more

167 jumps over the sampling interval. Option pricing under Merton’s model re-
168 quires integrating over the jump count distribution, but retains analytical
169 tractability through the characteristic function.

170 Kou [25] extended the jump-diffusion framework by replacing the Gaus-
171 sian jump amplitude distribution with a double-exponential (asymmetric
172 Laplace) distribution, which provides a better fit to the empirically observed
173 asymmetry between the sizes of positive and negative jumps. This modi-
174 fication preserves analytical tractability for option pricing while improving
175 the empirical fit to return distributions. Jump-diffusion models are partic-
176 ularly relevant in the context of equity markets, where identifiable macroe-
177 conomic announcements, earnings surprises, and geopolitical events generate
178 discrete, large-magnitude price movements that are distinct from the con-
179 tinuous diffusion component [18]. A fundamental challenge, however, is that
180 jump-diffusion models in their standard form do not provide an explicit mech-
181 anism for volatility persistence; the post-jump volatility immediately returns
182 to its baseline level, absent a separate stochastic volatility component. Inte-
183 grating jump-diffusion with regime-switching provides a natural solution to
184 this limitation, motivating the hybrid framework developed in the present
185 study.

186 *Hidden Markov models in finance*

187 Hidden Markov models provide a probabilistic framework for non-stationary
188 time series in which the observed data are generated by a process that
189 switches among a discrete set of latent states according to Markov dynam-
190 ics. The foundational algorithmic framework was developed by Rabiner
191 and Juang [15], who introduced the Baum-Welch algorithm for maximum-
192 likelihood estimation of the emission and transition parameters from observed
193 sequences. Hamilton [26] applied the regime-switching HMM to macroeco-
194 nomic time series, demonstrating that U.S. GDP growth is well described by
195 a two-state model with distinct expansion and recession regimes; this work
196 established the template for applying latent-state models to financial time
197 series. The identification of time-varying market regimes has since become a
198 central application of HMMs in finance, with applications ranging from stock
199 return modeling to volatility forecasting and asset allocation.

200 Rydén, Teräsvirta, and Åsbrink [16] provided an influential systematic
201 evaluation of HMM specifications against the stylized facts of daily return se-
202 ries. They showed that simple Gaussian-emission HMMs with a small number

203 of states could reproduce leptokurtosis and the absence of return autocorrela-
 204 tion, but failed to generate the observed volatility clustering; ACF of absolute
 205 returns under standard HMMs decayed far too rapidly relative to empirical
 206 data. This diagnostic limitation identified the core structural gap that the
 207 present study addresses through the jump-duration extension. Bulla and
 208 Bulla [17] advanced this line of inquiry by considering hidden semi-Markov
 209 models, in which the dwell time in each state follows a state-specific distri-
 210 bution rather than the geometric distribution implied by standard Markov
 211 dynamics. They showed that the semi-Markov extension substantially im-
 212 proves the reproduction of volatility clustering by allowing the model to dwell
 213 in high-variance states for durations drawn from a flexible parametric fam-
 214 ily. Rossi and Gallo [27] applied HMMs to multivariate volatility estimation,
 215 demonstrating that latent state dynamics can capture the joint clustering
 216 of volatility across assets in a more interpretable manner than multivariate
 217 GARCH models. Nguyen [28] further demonstrated the utility of HMMs for
 218 stock trading, showing that regime identification from price data can inform
 219 tactical asset allocation.

220 A persistent limitation of standard HMM approaches in finance is the
 221 computational burden of maximum-likelihood estimation via the EM algo-
 222 rithm, particularly when the number of states is large or the asset universe
 223 is broad. The Baum-Welch procedure requires iterative forward-backward
 224 passes through the data and may converge to local maxima depending on
 225 initialization. For high-dimensional or large-state-space models, this creates
 226 a barrier to scalable application. The present study addresses this limita-
 227 tion directly by replacing EM with direct frequentist counting of transitions
 228 between quantile-defined states, an approach that is computationally trivial
 229 and free of initialization sensitivity.

230 *Synthetic data generation and generative frameworks*

231 The generation of synthetic financial time series has attracted increasing
 232 attention as both a practical tool—for augmenting limited data, testing risk
 233 models, and stress testing portfolios—and as a benchmark for evaluating the
 234 quality of generative models. Assefa et al. [1] provide a comprehensive survey
 235 of the opportunities, challenges, and pitfalls of synthetic data generation in
 236 finance, noting that stylized-facts reproduction is the primary criterion for
 237 evaluating the statistical quality of synthetic financial time series, and that
 238 naive generative approaches tend to fail on at least one of the three core
 239 stylized facts.

240 The rise of generative adversarial networks (GANs) prompted several
 241 investigations into whether deep learning architectures could learn stylized
 242 facts from financial data. Takahashi, Chen, and Tanaka-Ishii [12] showed that
 243 standard GAN architectures produce synthetic return sequences that visually
 244 resemble financial time series but fail to reproduce ACF structure precisely.
 245 Kwon and Lee [13] provided a careful evaluation of GAN-based methods
 246 against the stylized-facts checklist, finding that while GANs match marginal
 247 distributions reasonably well, capturing temporal dependence structures—
 248 particularly volatility clustering—remains challenging without explicitly en-
 249 coding the relevant inductive biases into the architecture or training ob-
 250 jective. Foundation models for Markov jump processes, as introduced by
 251 Berghaus et al. [29], offer a complementary perspective: large-scale pre-
 252 trained inference networks can be applied to new Markov process instances
 253 without retraining, but their application to financial time series generation
 254 remains an open research direction.

255 Against this backdrop, the hybrid HMM framework of the present study
 256 occupies a distinct position: it is explicitly designed around the structural
 257 properties known to generate stylized facts, uses an interpretable discrete-
 258 state representation that can be audited and stress-tested, and achieves
 259 strong statistical fidelity with minimal computational overhead. The empiri-
 260 cal estimation procedure avoids the training instability and data requirements
 261 of deep generative models, while the jump-duration mechanism provides a
 262 principled solution to the volatility-clustering limitation of standard HMMs.
 263 These properties make the framework particularly well-suited for deployment
 264 in risk management and stress-testing pipelines, where interpretability and
 265 computational reproducibility are as important as statistical fidelity.

266 *Factor models and multi-asset extensions*

267 Modeling the joint dynamics of large asset universes requires a framework
 268 for managing cross-sectional dependence. The Single-Index Model (SIM) of
 269 Sharpe [19] provides the canonical linear factor decomposition: each asset’s
 270 excess return is decomposed into a common market factor component and an
 271 idiosyncratic residual, drastically reducing the number of parameters required
 272 to specify a covariance matrix relative to a full multivariate model. The SIM
 273 remains foundational in portfolio construction and risk management despite
 274 the availability of richer multi-factor models such as Fama-French [30], be-
 275 cause its simplicity enables transparent attribution and scalable computation.

Existing HMM-based work in finance has been predominantly univariate, focusing on single-asset or single-index regime identification. Applications to individual equities [28], volatility modeling [27], and macro time series [26, 31] collectively demonstrate the utility of latent state representations, but leave open the question of how HMM-derived generative models can be extended to large cross-sections of assets in a computationally tractable way. Mixture hidden Markov models [32] offer a partial solution by allowing multiple assets to share a common latent state structure, but estimating such models jointly across hundreds of assets is computationally demanding.

The present study addresses this gap by combining the univariate HMM framework with the SIM factor decomposition. The hybrid HMM is estimated on the SPY index, which serves as the market-factor generator; asset-level paths are then reconstructed via the linear SIM projection. This construction exploits the fact that the dominant source of joint volatility clustering in equity markets is the common factor, and that idiosyncratic shocks can be treated as approximately independent given the factor. The result is a scalable generative pipeline for a 424-asset universe that inherits the stylized-facts fidelity of the HMM without requiring joint estimation of a high-dimensional model.

295 Methods

296 *Data.*

297 The empirical foundation of this study was a dataset composed of 424
 298 United States-listed equities and exchange-traded funds spanning 10 years
 299 (2014-2024). A pipeline was developed to support the automated construc-
 300 tion of excess growth rate models for any ticker within this dataset, allowing
 301 for batch simulation and stress testing across market sectors. To validate
 302 the performance of this pipeline, we focused our analysis on the single-asset
 303 SPY, which tracked the Standard & Poor’s (S&P) 500 index. The data in-
 304 cluded daily open, high, low, and close prices along with volume-weighted
 305 average price metrics (VWAP). For training purposes, the first 2,766 obser-
 306 vations constituted the in-sample dataset, spanning from January 3, 2014,
 307 to December 31, 2024. A subsequent 249 trading days, from January 2 to
 308 December 31, 2025, were reserved for out-of-sample testing to assess model
 309 generalizability.

310 *Excess growth rate calculation.*

311 We computed the excess growth rate $G_{i,j}$ for ticker i between time pe-
 312 riod $j - 1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share
 313 (USD/share)). To isolate the stochastic premium earned over a risk-free
 314 asset, we computed the excess growth rate for ticker i during the period
 315 $j - 1 \rightarrow j$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

316 where Δt represented the time step set to 1/252 for daily data (units: years),
 317 and r_f denotes the constant continuously compounded risk-free rate (units:
 318 year⁻¹) derived from Separate Trading of Registered Interest and Principal
 319 of Securities (STRIPS) bond yields observed on September 10, 2025. Un-
 320 like return, the excess growth rate has units of inverse time (year⁻¹) and
 321 directly quantifies the premium earned above a continuously compounded
 322 risk-free benchmark. Log growth rates are time-additive and consistent with
 323 a continuous compounding framework. While r_f varied dynamically over the
 324 sample period, a constant proxy was employed here to isolate the endogenous
 325 volatility dynamics without introducing exogenous rate-curve noise.

326 *Discrete hidden Markov model.*

327 We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$
 328 [15]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood
 329 regimes and was defined by quantile bins of a fitted cumulative distribu-
 330 tion function (CDF) for the excess growth rate series. The observation space
 331 $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition
 332 matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was
 333 parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the
 334 stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \quad (2)$$

335 To map observations to hidden states during estimation, we defined quantile
 336 boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N - 1\}$ and used finite
 337 outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. An
 338 observation was assigned to state k when $Q_{k-1} < R_t \leq Q_k$.

339 The choice of the Laplace distribution over alternatives such as the Stu-
 340 dent’s t-distribution was motivated by its alignment with the peaks and
 341 heavy tails of high-frequency financial excess growth rates [4, 33]. While the
 342 Student’s t-distribution was often used to model leptokurtosis, the Laplace
 343 distribution’s peak at the mean matched the concentration of small price
 344 movements observed in the SPY dataset [3, 34]. Furthermore, the Laplace
 345 distribution provided a closed-form for quantile estimation, supporting the
 346 424-ticker pipeline. While maximum likelihood estimation (MLE) for the
 347 Student’s t-distribution often required iterative numerical optimization, the
 348 Laplace distribution allowed for a direct MLE approach [35]. The quantile
 349 boundaries were derived from a parametric Laplace fit and therefore did not
 350 enforce equal counts per bin in the historical data; instead, they provided a
 351 partition of the distribution that preserved tail shape. By using the Laplace
 352 distribution to inform the binning process, the model used this baseline to
 353 characterize the peak and tails of the excess growth rate series before the
 354 empirical transition logic and jump-diffusion mechanism were applied [25].

355 We estimated the transition matrix $\mathbf{T}$ through direct frequentist count-
 356 ing rather than iterative optimization. This approach differed from the tra-
 357 ditional expectation-maximization (EM) algorithm used for parameter esti-
 358 mation in Gaussian mixture and hidden Markov models [35]. However, this
 359 approach was justified because the discrete state assignments allowed for
 360 straightforward counting of observed transitions between regimes, and it was
 361 conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [36]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k , the continuous excess growth rate ($\hat{R}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model preserved the discontinuous components of the price process important for asset modeling while maintaining a smooth representation of intra-state variance [25].

Growth rate generation.

Let's outline a simple sampling algorithm for generating random sequences from an HMM (without jumps). We initialize the generation process by drawing an initial hidden state X_0 from the stationary distribution $\bar{\pi}$. The stationary distribution $\bar{\pi}$ represented the long-run probability of the market occupying various regimes or moods [37]. While $\bar{\pi}$ mathematically satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$, we estimated this distribution numerically by raising the transition matrix $\mathbf{T}^n$ to a sufficiently large power $n = 50$, ensuring the vector reflected the equilibrium of market dynamics [38].

Given an HMM defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E}, \bar{\pi})$, the number of time steps T , and the stationary distribution $\bar{\pi}$. Draw an initial hidden state X_0 from the initial distribution: $X_0 \sim \bar{\pi}$. For $n = 1$ to T , we sample the next hidden state X_n from the `categorical` distribution $\mathbb{P}_H(\dots)$ defined by the previous state X_{n-1} and the transition matrix $\mathbf{T}$: $X_n \sim \mathbb{P}_H(X_n | X_{n-1})$. Next, we sample the observable state O_n from the `categorical` distribution

399 $\mathbb{P}_E(\dots)$ defined by the current hidden state X_n and the emission matrix $\mathbf{E}$:
 400 $O_n \sim \mathbb{P}_E(O_n \mid X_n)$. Finally, we store the sequences of hidden states and
 401 observable states. This procedure generates a sequences of hidden states
 402 $\{X_1, X_2, \dots, X_T\}$ and observable states $\{O_1, O_2, \dots, O_T\}$. The initial state
 403 X_0 is used to generate the sequence but is not included in the output.

404 *Hyperparameter optimization via grid search.*

405 To rigorously identify the optimal jump probability (ϵ) and mean jump
 406 duration (λ) [9, 25], we implemented a multi-objective grid search designed
 407 to minimize the discrepancy between the historical and simulated market sig-
 408 natures. The error function specifically targeted two stylized facts prevalent
 409 in financial time series [4, 16]: volatility clustering and leptokurtosis. We
 410 measured volatility clustering using the sum of squared errors between the
 411 observed and simulated autocorrelation function (ACF) of absolute excess
 412 growth rates [22] up to a lag of $L = 252$ trading days (approximately one
 413 trading year). To preserve the heavy-tailed nature of the distribution ini-
 414 tially noted in foundational market studies [3], we included a penalty term
 415 for the global kurtosis.

416 The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L (\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau))^2 + w_K (K_{obs} - \overline{K}_{sim})^2 \quad (3)$$

417 where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth au-
 418 tocorrelation at lag τ and the global kurtosis, respectively. The simulated
 419 equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 200 independent syn-
 420 thetic paths of 2,766 trading days per grid point [36]. A penalty weight of
 421 $w_K = 0.20$ was assigned to prioritize tail behavior alongside the temporal
 422 ACF decay. The grid search systematically explored ϵ values from 10^{-4} to
 423 2.5×10^{-2} and λ values from 10 to 160. The full computational procedure is
 424 detailed in Algorithm 4.

425 *Statistical validation metrics*

426 To assess whether synthetic paths are statistically consistent with empiri-
 427 cal data, we applied two nonparametric goodness-of-fit tests to the marginal
 428 distributions of simulated excess growth rate sequences. For each of the 1,000
 429 simulated paths, the empirical cumulative distribution function (CDF) of the

430 simulated sequence was compared against the empirical CDF of the historical
 431 in-sample or out-of-sample observations.

432 The Kolmogorov-Smirnov (KS) statistic is defined as:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \quad (4)$$

433 where F_n and F_m denote the empirical CDFs of the observed and simulated
 434 sequences of lengths n and m , respectively [39, 40]. The null hypothesis of
 435 distributional equivalence is rejected at significance level α when D_{KS} exceeds
 436 the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

437 The Anderson-Darling (AD) statistic places greater weight on the distri-
 438 butional tails than the KS test and is defined as:

$$A^2 = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln F(X_{(i)}) + \ln (1 - F(X_{(n+1-i)}))] \quad (5)$$

439 where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics of the pooled sample and
 440 F is the empirical CDF of the reference distribution [41]. The AD test
 441 is particularly suited for assessing tail-distributional fidelity, which is the
 442 primary concern for risk management applications.

443 We report the *pass rate* as the proportion of the 1,000 simulated paths for
 444 which the null hypothesis of distributional equivalence was not rejected at
 445 the $\alpha = 0.05$ significance level. A pass rate above 95 percent indicates that
 446 the synthetic paths are, in aggregate, statistically indistinguishable from the
 447 historical data under that test. Pass rates are reported separately for the
 448 in-sample and out-of-sample windows and for both model variants (HMM
 449 without jumps and the hybrid HMM with jumps).

450 *Single-Index Factor Model extension*

451 The framework described above applies to any individual asset ticker
 452 within the 424-asset dataset. To generalize synthetic path generation to a
 453 correlated multi-asset setting without fitting a separate HMM for each asset,
 454 we exploit the linear factor structure of the Single-Index Model (SIM) [19].
 455 The SIM expresses the excess growth rate of asset i at time t as:

$$G_{i,t} = \alpha_i + \beta_i \cdot G_{SPY,t} + \eta_{i,t}, \quad \eta_{i,t} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\eta,i}^2) \quad (6)$$

456 where $G_{SPY,t}$ denotes the SPY excess growth rate at time t , β_i measures the
 457 systematic sensitivity of asset i to the index, α_i is the idiosyncratic drift, and

458 $\eta_{i,t}$ is a zero-mean idiosyncratic shock assumed independent across assets and
 459 time. The parameters $(\alpha_i, \beta_i, \sigma_{\eta,i}^2)$ are estimated via ordinary least squares
 460 on the in-sample time series for each asset i .

461 Synthetic multi-asset paths are generated as follows. First, the hybrid
 462 HMM produces a single synthetic SPY excess growth rate path $\{\hat{G}_{\text{SPY},t}\}_{t=1}^T$
 463 using the regime-switching and jump-duration mechanism described above.
 464 Second, for each asset i , an idiosyncratic shock sequence $\{\hat{\eta}_{i,t}\}$ is drawn
 465 independently from $\mathcal{N}(0, \hat{\sigma}_{\eta,i}^2)$. Third, the synthetic excess growth rate for
 466 asset i is recovered via:

$$\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{\text{SPY},t} + \hat{\eta}_{i,t}. \quad (7)$$

467 This construction preserves the cross-sectional correlation structure induced
 468 by the common factor while maintaining the regime-switching and jump-
 469 diffusion dynamics of the index. The resulting 424-dimensional synthetic
 470 path can be used for stress testing, portfolio risk assessment, and scenario
 471 generation across the full asset universe in a single generative pass, avoiding
 472 the curse of dimensionality that would accompany a full multivariate HMM.

473 *Sensitivity analysis: low-resolution models*

474 This section provides the results for lower state resolutions ($N = 30$ and
 475 $N = 60$) to demonstrate how state granularity impacts the model’s ability to
 476 capture stylized facts. Table 3 summarizes the pass rates for the Kolmogorov-
 477 Smirnov and Anderson-Darling tests across these four cases. As the number
 478 of states decreases, the model maintained overall distribution fit, though tail-
 479 region precision decreased slightly compared to the 90-state configuration.

480 Empirical Study

481 We computed descriptive statistics and stylized-facts tests for the SPY
 482 daily excess growth rate series over the in-sample (2014–2024, $T = 2,766$) and
 483 out-of-sample (2025, $T = 249$) windows. We observed excess kurtosis of 7.7
 484 (IS) and 6.9 (OoS), rejection of the Jarque-Bera normality test ($p < 0.001$)
 485 in both windows, and rejection of the Ljung-Box test on $|G_t|$ ($p < 0.001$)
 486 in both windows, which confirms that the three canonical stylized facts—
 487 leptokurtosis, negligible raw-return autocorrelation, and persistent volatility
 488 clustering—are present in the target data and must be reproduced by any
 489 faithful generative model (Table 1).

490 We applied the multi-objective grid search to identify the optimal jump
 491 hyperparameters over the search grid $\epsilon \in [10^{-4}, 2.5 \times 10^{-2}]$ and $\lambda \in [10, 160]$,
 492 evaluating 200 simulated paths per grid point. We found a well-defined in-
 493 terior minimum at $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$, which indicates that the jump
 494 hyperparameters are identifiable from data and that the objective landscape
 495 is not degenerate (Figure 5, panel a). We compared the ACF of $|G_t|$ at the
 496 optimal parameters against the observed SPY ACF and observed close agree-
 497 ment across lags 1–252, which confirms that the optimized jump mechanism
 498 reproduces the empirical ARCH effect (Figure 5, panel b).

499 We selected a state resolution of $N = 100$ for the primary analysis and ex-
 500 amined the sensitivity of all quality metrics to $N \in \{30, 60, 90, 100, 150, 200\}$.
 501 We observed that KS and AD pass rates remain stable across this range
 502 for both HMM-NJ ($\geq 98.9\%$ KS, $\geq 97.9\%$ AD) and HMM-WJ ($\geq 96.0\%$ KS,
 503 $\geq 88.6\%$ AD), and that ACF-MAE for HMM-WJ improves slightly from 0.057
 504 at $N = 30$ to 0.049 at $N = 200$, which suggests that the framework is robust
 505 to the choice of state resolution over a wide range (Table 3). We verified
 506 via state-occupancy diagnostics that all states at $N \leq 200$ contain at least
 507 7 observations, which confirms that the Laplace equal-probability partition
 508 maintains adequate statistical support per state. At $N = 350$ certain states
 509 are never visited in the historical record, leaving zero transition probabilities
 510 that break the ergodic structure of the chain, which establishes the practical
 511 upper bound on state resolution.

512 We simulated 1,000 paths of 2,766 trading days from each of three models—
 513 GARCH(1,1), HMM-NJ, and HMM-WJ—and applied two-sample Kolmogorov-
 514 Smirnov and Anderson-Darling tests at $\alpha = 0.05$ to assess distributional con-
 515 sistency with the historical data. We found that GARCH(1,1) achieved IS
 516 KS and AD pass rates of only 4.6% (SE 0.7) and 1.1% (SE 0.3), which indi-

517 cates that the GARCH marginal distribution is systematically inconsistent
518 with the observed data despite its well-known volatility clustering proper-
519 ties (Table 2). We found that HMM-NJ achieved IS pass rates of 99.4%
520 (SE 0.2) and 98.5% (SE 0.4), and HMM-WJ achieved 97.1% (SE 0.5) and
521 90.4% (SE 0.9), which confirms that the Laplace-mixture emission structure
522 adequately captures the observed heavy-tailed distribution (Table 2).

523 We computed the ACF of $|G_t|$ over lags 1–252 for each model and com-
524 pared it against the empirical ACF. We observed IS ACF-MAE values of
525 0.031 for GARCH, 0.059 for HMM-NJ, and 0.052 for HMM-WJ. We found
526 that GARCH achieves the lowest ACF-MAE because its variance equation
527 is explicitly designed to match autocorrelation structure, yet this comes at
528 the cost of distributional fidelity—a tradeoff that the quality metrics expose
529 directly (Table 2). We observed that HMM-NJ produces an ACF of $|G_t|$
530 that decays rapidly toward zero rather than exhibiting the empirically ob-
531 served slow decay, which confirms that the standard transition matrix alone
532 is structurally incapable of generating persistent volatility clustering (Fig-
533 ure 4, panel b). We observed that HMM-WJ generates a mixture of two
534 path families: approximately 76% of IS paths contain no Poisson jump and
535 behave identically to HMM-NJ, while the remaining $\sim 24\%$ contain at least
536 one jump event and sustain substantial ACF($|G_t|$) decay across all lags, which
537 demonstrates that the jump-duration mechanism is the structural feature re-
538 sponsible for reproducing the ARCH effect (Figure 4, panel b).

539 We evaluated out-of-sample performance over 249 trading days (Jan-
540 uary 2–December 31, 2025) using the same 1,000-path ensemble. We found
541 that GARCH achieved OoS KS and AD pass rates of 79.0% (SE 1.3) and
542 71.2% (SE 1.4), while HMM-NJ achieved 96.7% (SE 0.6) and 96.3% (SE 0.6),
543 and HMM-WJ achieved 95.0% (SE 0.7) and 95.4% (SE 0.7), which shows that
544 the HMM variants generalise robustly while GARCH degrades substantially
545 (Table 2). We observed that the simulated kurtosis for GARCH collapsed to
546 1.7 out-of-sample against an observed value of 6.9, while the HMM variants
547 maintained simulated kurtosis near 5.0–5.1, which suggests that GARCH
548 does not preserve tail behaviour outside the training window (Table 2). We
549 observed that the distribution of KS p -values shifted leftward from in-sample
550 to out-of-sample, which reflects the expected distributional degradation when
551 stationary parameters are applied to a period with elevated macro uncer-
552 tainty (Figure 6, panels a–b). We found that the observed ACF of $|G_t|$ in
553 the 2025 test window exceeded the HMM-WJ simulation band at medium-
554 to-long lags, which indicates that the IS-calibrated jump hyperparameters

555 (ϵ^*, λ^*) underestimate the frequency and persistence of volatility episodes in
 556 this period and that the jump parameters are regime-dependent (Figure 6,
 557 panel d). Despite this ACF gap, the observed marginal density remained
 558 within the simulation envelope, which confirms adequate distributional cov-
 559 erage even under regime shift (Figure 6, panel c).

560 We propagated HMM-WJ SPY paths through a Single-Index Model for
 561 424 S&P 500 constituents, decomposing each asset’s excess growth rate as
 562 $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$, where $(\hat{\alpha}_i, \hat{\beta}_i)$ were estimated by ordinary least
 563 squares on the 2014–2024 training window and the idiosyncratic shocks $\hat{\eta}_{i,t}$
 564 were drawn by resampling the empirical residuals. We observed that the SIM
 565 R_i^2 varied widely across GICS sectors, from near-zero for defensive and low-
 566 correlation names to above 0.8 for broad-market ETFs, which confirms that
 567 the single-factor structure captures only the systematic component of each
 568 asset’s dynamics (Figure 7, panel a; Table 4). We found IS KS pass rates
 569 with a median of 66.7% and a mean of 58.4% across the 424-asset universe,
 570 which are lower than the 97.1% achieved for SPY alone and which reflect the
 571 well-known limitation that the SIM’s single-factor linearity and symmetric
 572 residual distribution cannot fully capture asset-specific tail behaviour, skew-
 573 ness, or sector-level dynamics (Figure 7, panel b; Table 4). We observed
 574 that the degradation in KS pass rate was concentrated among assets with
 575 low R_i^2 —those dominated by idiosyncratic risk that the single factor cannot
 576 reach—while high- β assets closely tracking SPY retained pass rates above
 577 80%, which suggests that the HMM-WJ generative engine is not the bottle-
 578 neck and that the limiting factor is the expressiveness of the factor model
 579 (Figure 7, panel c). We found OoS KS pass rates with a mean of 82.1% and
 580 a median of 91.8% across 417 surviving assets, which shows that the SIM
 581 extension generalises to the out-of-sample window (Table 4).

582 We computed the correlation-distance distribution between the out-of-
 583 sample simulated paths and the historical trajectory. We observed a mean
 584 distance of $1 - \rho \approx 1.0$, which confirms that the generative framework pro-
 585 duces temporally independent realisations that do not attempt to predict
 586 specific historical events, distinguishing it from a forecasting tool and posi-
 587 tioning it as an instrument for stress testing and scenario generation (Fig-
 588 ure 6).

589 Discussion

590 This study introduced a hybrid hidden Markov modeling framework to
591 replicate the statistical regularities of empirical equity excess growth rates.
592 By integrating a quantile-discretized state space with a Poisson-driven jump-
593 duration mechanism, we developed a generative framework that captured
594 the leptokurtic distributions and volatility clustering observed in the SPY
595 exchange-traded fund. Our results confirmed that while increasing the state
596 resolution to $N = 100$ provided the granularity to characterize the central
597 mass of the excess growth rate distribution, standard Markovian transitions
598 alone did not reproduce higher-order temporal dependencies. The inclusion
599 of discontinuous shocks through a jump-diffusion extension was needed to
600 drive the model into persistent high-volatility regimes, thereby replicating
601 the ARCH effect identified in historical data.

602 Statistical validation using Kolmogorov-Smirnov and Anderson-Darling
603 tests demonstrated high in-sample consistency, with pass rates of 97.1 per-
604 cent (SE 0.5) and 90.4 percent (SE 0.9), respectively, across 1,000 simulated
605 paths. Out-of-sample testing on 249 trading days from 2025 yielded pass
606 rates of 95.0 percent (SE 0.7) and 95.4 percent (SE 0.7), respectively. Fur-
607 thermore, the correlation distance analysis confirmed that our framework
608 produced independent realizations that remained uncorrelated with specific
609 historical timing; this satisfied the requirements of the efficient markets hy-
610 pothesis and distinguished our generative methodology from predictive fore-
611 casting tools [1, 13, 21]. The proposed framework offered an alternative to
612 standard likelihood-based estimation methods by relying on empirical es-
613 timation. By bypassing the complexities of the expectation-maximization
614 algorithm through direct frequentist counting, the model remained suited for
615 large-scale stress testing and risk management applications [36, 42, 43]. Fu-
616 ture research could extend this discrete-state logic to multi-asset portfolios
617 or incorporate foundation models for inference to further refine the detection
618 of Markov jump processes in higher-frequency data [29, 32]. Ultimately, the
619 hybrid hidden Markov model served as a foundation for simulating market
620 dynamics and generating synthetic time series that remained statistically
621 consistent with underlying market regimes [28, 37].

622 While the current framework was able to generate statistically consistent
623 excess growth rate distributions, there were several limitations that could be
624 addressed in future work. First, the model assumed stationarity in transi-
625 tion dynamics and jump hyperparameters, which may not hold during peri-

ods of structural market change. The out-of-sample ACF analysis (Figure 6,
 panel d) provided direct evidence of this limitation: the IS-calibrated (ϵ^*, λ^*)
 underestimated volatility clustering persistence in the 2025 test window, con-
 sistent with the elevated macro uncertainty of that period. Future extensions
 could incorporate time-varying transition matrices or regime-dependent jump
 parameters—for example, allowing ϵ and λ to evolve as functions of a latent
 volatility state or observable macro indicators—to capture evolving market
 conditions. Second, the use of a fixed quantile-based state space may limit
 the model’s ability to adapt to changing volatility regimes; adaptive bin-
 ning strategies or non-parametric state definitions could enhance flexibility.
 Third, while the Laplace distribution provided a good fit for excess growth
 rates, alternative distributions or mixture models could be explored to better
 capture tail behavior and skewness. Finally, the Single-Index Model exten-
 sion presented in Section ?? demonstrates that the HMM-WJ framework can,
 in principle, scale from a single asset to a full equity universe by coupling
 the generative SPY engine with an asset-level factor model. The SIM serves
 this role as a proof of concept: it is intentionally minimal (one factor, linear,
 symmetric residuals) and consequently inherits well-known limitations—the
 median in-sample KS pass rate of 66.7% across 424 assets confirms that a
 single-factor linear decomposition cannot capture the heterogeneous tail be-
 haviour, skewness, and sector-specific dynamics present in individual equities
 [19, 30]. Importantly, however, the architecture is modular: the HMM-WJ
 generative engine for the market factor is decoupled from the factor model
 used to propagate paths to individual assets. Replacing the SIM with a multi-
 factor specification (e.g., Fama–French three- or five-factor models [30]), a
 principal-component-based factor structure, or a nonlinear model such as
 a neural network residual generator could substantially improve asset-level
 distributional fidelity without modifying the underlying Markov chain ma-
 chinery. The out-of-sample SIM results (median KS pass rate of 91.8% over
 249 trading days in 2025) provide a baseline for such extensions and suggest
 that even a rudimentary factor model, when paired with the HMM-WJ, can
 produce statistically plausible multi-asset scenarios suitable for portfolio-level
 stress testing and risk assessment.

659 Conclusion

660 This study developed and evaluated a hybrid hidden Markov model (HMM)
661 framework for generating high-fidelity synthetic equity excess growth rate
662 paths. The quality of the synthetic data was assessed against the three core
663 stylized facts that any faithful financial time series generator must repro-
664 duce: leptokurtic marginal distributions, the absence of linear autocorrela-
665 tion in raw returns, and persistent volatility clustering. We summarize the
666 contributions below.

667 **Hybrid HMM with Poisson jump-duration mechanism.** The cen-
668 tral methodological contribution is the augmentation of a discrete-state regime-
669 switching model with a Poisson-driven jump-duration mechanism. Standard
670 Markov transition logic underproduces volatility persistence because empiri-
671 cally estimated transition matrices assign negligible probability to transitions
672 from central states into tail states, and even the tail states themselves exit
673 within one to two steps on average under pure Markovian dynamics. By
674 introducing a jump probability ϵ and mean jump duration λ drawn from a
675 Poisson distribution, the framework overrides standard transitions during pe-
676 riods of market stress and forces the model to dwell in high-volatility regimes
677 for realistic durations. This mechanism is minimal in parameterization—
678 requiring only two scalar hyperparameters—yet produces a qualitative shift
679 in the temporal correlation structure of absolute returns, as confirmed by
680 ACF comparison across lags up to 252 trading days.

681 **Empirical, non-iterative parameter estimation.** A second contri-
682 bution is the avoidance of the expectation-maximization (EM) algorithm in
683 favor of direct frequentist transition counting. In the standard HMM set-
684 ting, EM is used because latent state assignments are unobserved. Here, the
685 quantile-based state partition explicitly assigns each historical observation
686 to a hidden state, making direct counting possible. This simplification re-
687 duces computational cost, eliminates convergence sensitivity to initialization,
688 and produces a model that is transparent and reproducible—properties that
689 matter for large-scale pipelines operating over 424 assets.

690 **In-sample and out-of-sample validation.** Using SPY (2014–2024)
691 as a case study, the hybrid model achieved in-sample Kolmogorov-Smirnov
692 and Anderson-Darling pass rates of 97.1 percent (SE 0.5) and 90.4 percent
693 (SE 0.9), respectively, across 1,000 simulated paths. Out-of-sample testing
694 on 249 trading days from 2025 showed pass rates of 95.0 percent (SE 0.7)
695 and 95.4 percent (SE 0.7), respectively. The correlation-distance analysis

696 further confirmed that simulated paths are temporally independent of the
697 historical trajectory, consistent with the efficient markets hypothesis and the
698 requirements of a generative—rather than predictive—framework.

699 **Single-Index Factor Model extension for multi-asset generaliza-**
700 **tion.** The framework extends naturally to the full 424-asset universe via a
701 Single-Index Model (SIM) factor structure. By regressing each asset’s ex-
702 cess growth rate on the SPY factor and estimating idiosyncratic residual
703 volatility, synthetic correlated multi-asset paths can be generated in a sin-
704 gle pass: the hybrid HMM produces a common-factor path, and asset-level
705 paths are recovered by projecting through asset-specific loadings. This con-
706 struction preserves the regime-switching and jump-diffusion dynamics of the
707 index while maintaining the cross-sectional correlation structure, enabling
708 portfolio-level stress testing without fitting a high-dimensional multivariate
709 model.

710 *Broader implications*

711 The proposed framework offers practical utility both for risk manage-
712 ment and for the broader synthetic data community. Generative models
713 that reproduce the stylized facts of financial markets enable stress testing
714 under market regimes that are statistically plausible but not historically ob-
715 served, addressing a key limitation of scenario libraries derived solely from
716 empirical records. The interpretable regime structure—where each hidden
717 state corresponds to a quantile-defined segment of the excess growth rate
718 distribution—facilitates communication between quantitative analysts and
719 risk managers, as states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*)
720 and linked to economic narratives. The model’s computational efficiency
721 further supports Monte Carlo-based risk metrics such as Value-at-Risk and
722 Conditional Value-at-Risk at scale across large asset universes. More broadly,
723 the quality evaluation methodology demonstrated here—combining distribu-
724 tional fidelity tests (KS, AD) with temporal structure metrics (ACF-MAE)
725 and reporting standard errors across Monte Carlo ensembles—provides a
726 reusable template for assessing the quality of synthetic time series in other
727 domains where temporal dependence structures are critical to downstream
728 applications.

729 *Limitations*

730 Several limitations inform the interpretation of these results. First, the
731 model assumes stationarity of the transition matrix across the full in-sample

732 window; structural breaks in market microstructure—such as those accom-
 733 panying regulatory changes or the COVID-19 shock—may not be adequately
 734 captured by a single static transition matrix. Second, the quantile-based
 735 state partition is fixed at estimation time and does not adapt to evolving
 736 volatility regimes; the boundaries are calibrated on the full in-sample distri-
 737 bution and may overweight the central mass relative to the tails in periods
 738 of prolonged market stress. Third, the SIM factor structure imposes a lin-
 739 ear, single-factor model of cross-sectional dependence; this may understate
 740 tail co-movement during systemic events where correlations rise sharply and
 741 the Gaussian idiosyncratic shock assumption fails. Finally, the out-of-sample
 742 evaluation covers a single 249-day window (full calendar year 2025), which
 743 limits the statistical power of out-of-sample inference; a longer or rolling
 744 out-of-sample evaluation would provide a more robust assessment.

745 *Future work*

746 Several extensions follow naturally from the current framework. Time-
 747 varying transition matrices—estimated via rolling windows or Bayesian on-
 748 line learning—would allow the model to adapt to structural shifts in regime
 749 persistence. Multivariate HMMs, in which the hidden state jointly governs
 750 a small number of correlated assets, could capture higher-order dependen-
 751 cies beyond the single-factor SIM approximation and are particularly rele-
 752 vant for sector-level stress testing. Adaptive state definitions, such as non-
 753 parametric density-based clustering of excess growth rates, could replace the
 754 fixed Laplace quantile partition and improve tail coverage during crisis pe-
 755 riods. Alternative factor structures, including the Fama-French three-factor
 756 model [30], could enrich the SIM extension by decomposing idiosyncratic
 757 risk further across size and value dimensions. Finally, integrating the hybrid
 758 HMM as the generative component within a larger portfolio optimization
 759 loop—where the model produces synthetic scenarios used to estimate tail-risk
 760 objectives—would provide an end-to-end computational finance application
 761 of the framework.

762 **Conflict of Interest Statement**

763 The authors declare that the research was conducted without any com-
764 mercial or financial relationships that could potentially create a conflict of
765 interest.

766 **Author Contributions**

767 J.V. directed the study. A.A. developed the model and simulation code,
768 conducted the in-sample and out-of-sample analysis, and generated the fig-
769 ures. Both authors edited and reviewed the final manuscript.

770 **Data Availability Statement**

771 The model code, simulation scripts, training and testing data are avail-
772 able under a Massachusetts Institute of Technology (MIT) license from the
773 GitHub repository: [https://github.com/varnerlab/HMM-Model-with-Jumps.](https://github.com/varnerlab/HMM-Model-with-Jumps.git)
774 [git](https://github.com/varnerlab/HMM-Model-with-Jumps.git)

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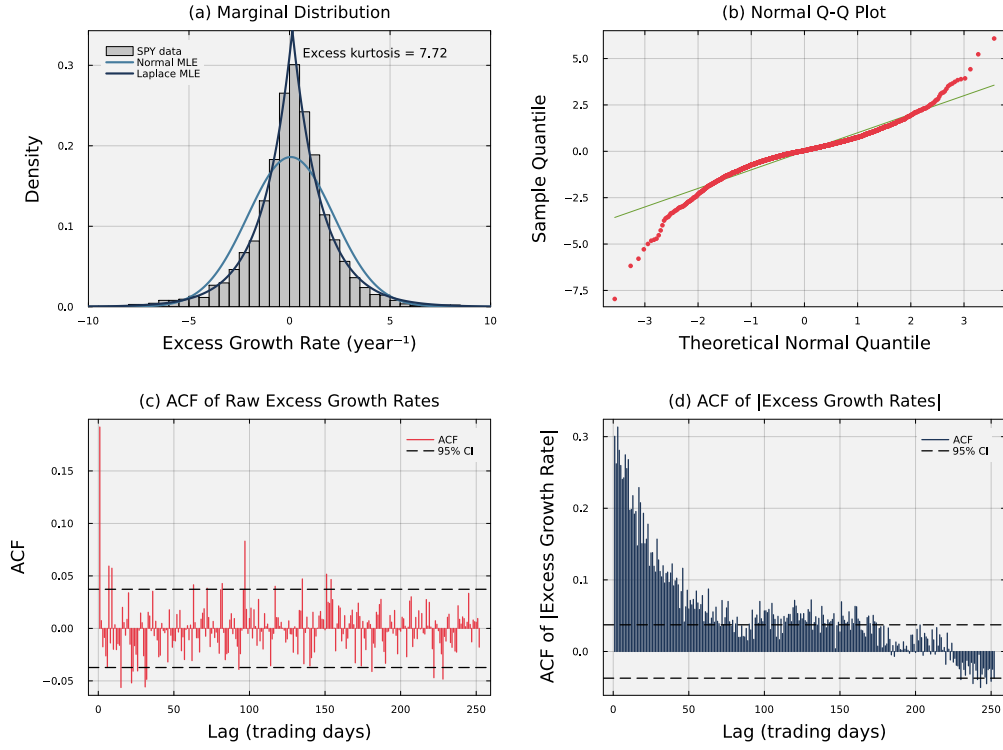


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns — consistent with the efficient markets hypothesis — against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

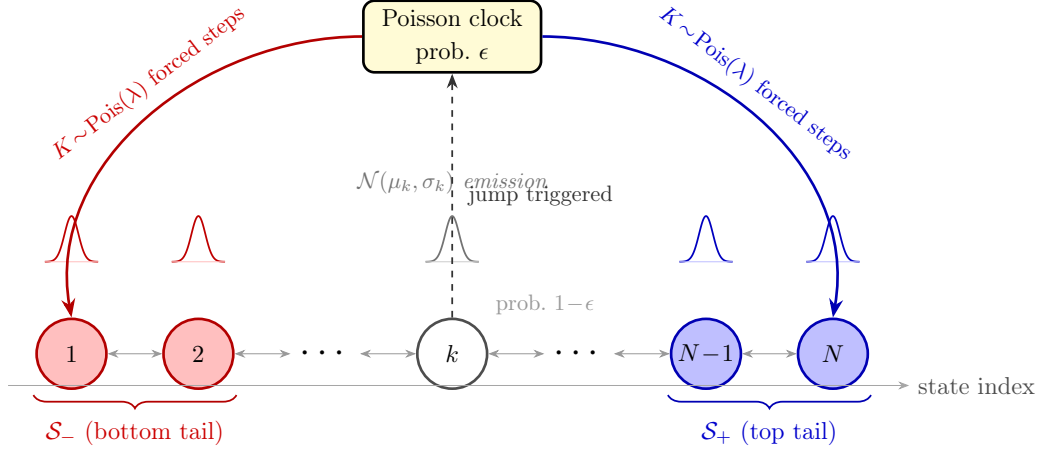


Figure 2: Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ , dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Pois}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional $\mathcal{N}(\mu_k, \sigma_k)$ emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

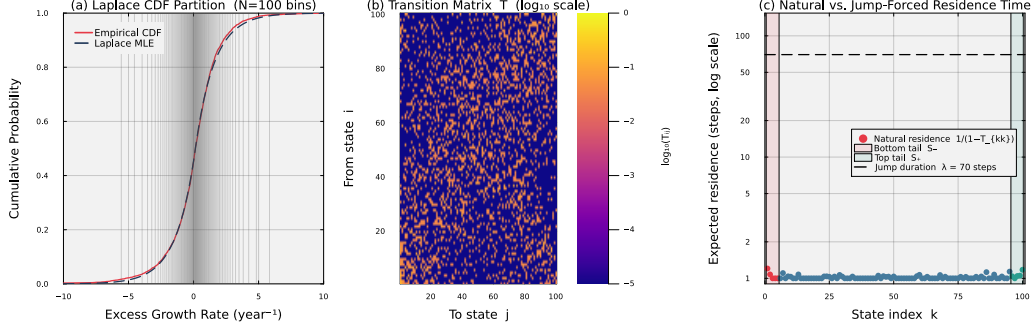


Figure 3: Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix T in $\log_{10}$ colour scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set S_- (states 1–5, red shading) and top tail set S_+ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean jump duration $\lambda = 70$ steps imposed by the Poisson jump mechanism, quantifying the two-order-of-magnitude gap that motivates the jump-duration extension.

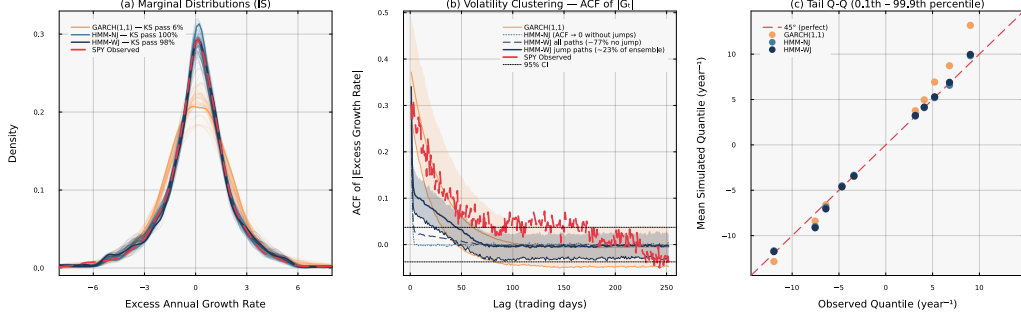


Figure 4: Head-to-head in-sample model comparison for SPY ($N = 100$, 1,000 simulated paths). Panel (a): marginal density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 94% of paths (pass rate 6%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at rates $\geq 96\%$, confirming that the Laplace-mixture emission adequately captures the observed heavy tails. HMM-WJ generalises robustly out-of-sample (OoS KS pass rate 95%), while GARCH recovers partially (OoS 78%). Panel (b): autocorrelation function of absolute excess growth rates, $\text{ACF}(|G_t|)$, at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so $\text{ACF}(|G_t|) \approx 0$ for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining $\sim 24\%$ of paths contain at least one jump event and sustain substantial ACF($|G_t|$) decay across all lags (solid navy curve $\pm$ band). The jump frequency—and hence the fraction of the ensemble exhibiting slow ACF decay—is controlled by the hyperparameter λ , making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

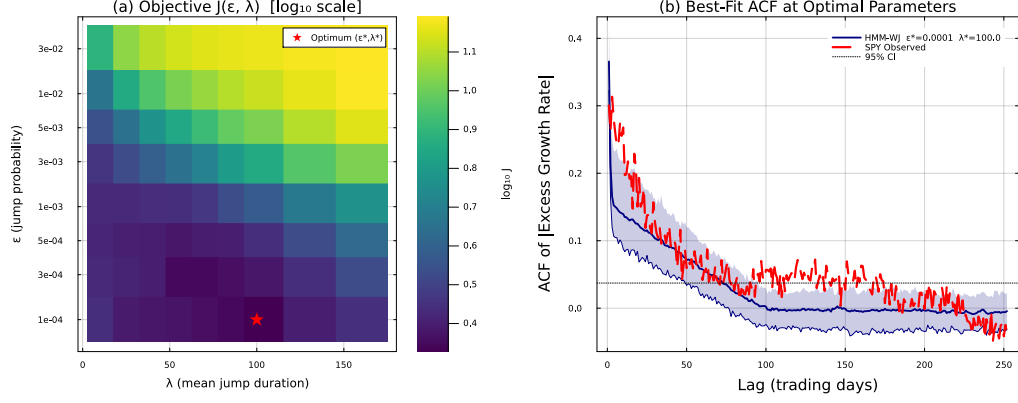


Figure 5: Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100, 200$ paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid; the red star marks the optimal pair (ϵ^*, λ^*) . The well-defined interior minimum confirms that the jump hyperparameters are identifiable from data. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).

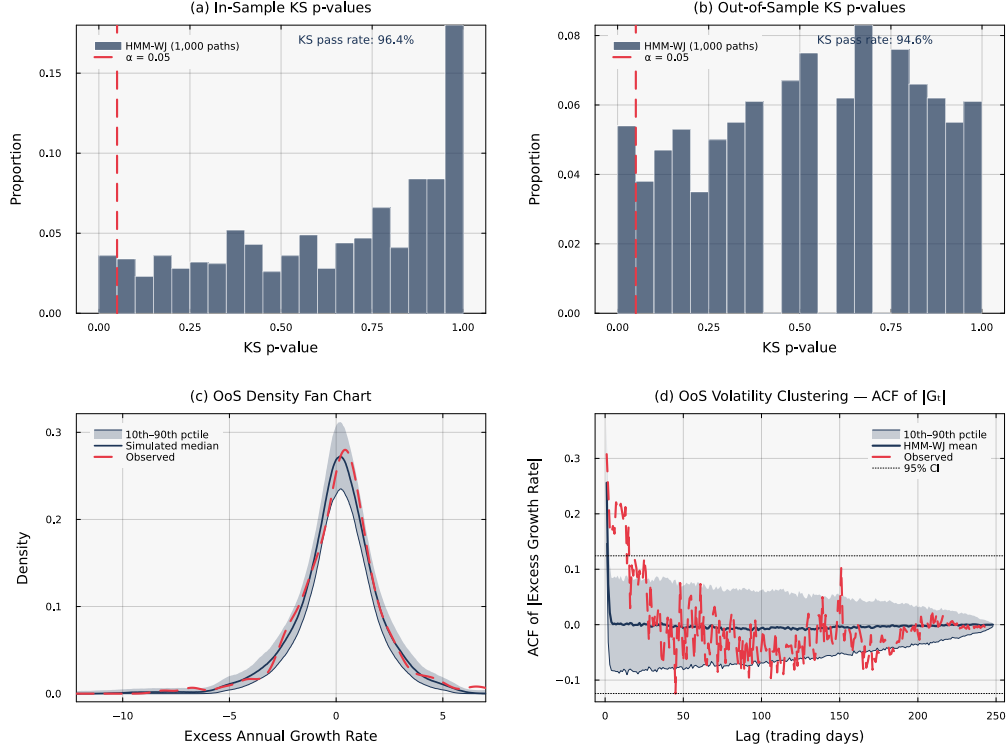


Figure 6: Statistical validation of HMM-WJ ($N = 100, 1,000$ simulated paths, $\alpha = 0.05$). Panels (a) and (b) show the distribution of two-sample KS p -values in-sample and out-of-sample, respectively; a well-calibrated generative model produces p -values that are approximately uniform above the significance threshold. The leftward shift from (a) to (b) reflects the expected distributional degradation when stationary parameters are applied to a regime with elevated macro uncertainty. Panel (c) shows the out-of-sample marginal density fan chart; the observed density (dashed red) falls within the 10th–90th percentile simulation envelope, confirming adequate distributional coverage despite the regime shift. Panel (d) shows the ACF of $|G_t|$ in the out-of-sample window. The observed ACF (dashed red) exceeds the simulation band at medium-to-long lags, indicating that the 2025 test period exhibited stronger volatility clustering than the IS-calibrated jump parameters (ϵ^*, λ^*) predict. This gap provides direct evidence that the jump hyperparameters are regime-dependent and motivates the time-varying extensions discussed in Section .

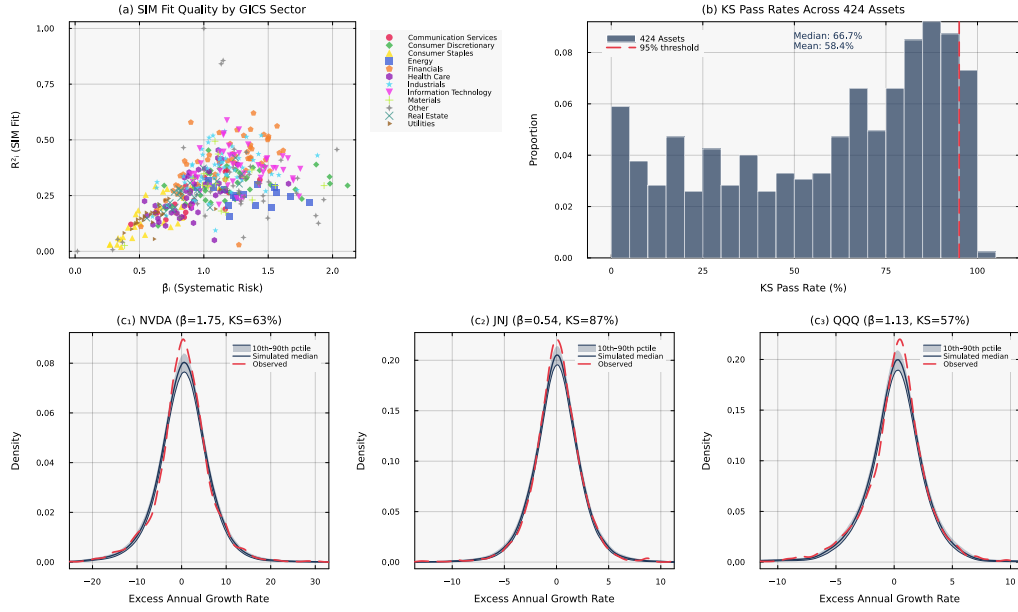


Figure 7: Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) summarises the SIM regression fit quality by GICS sector. Panel (b) shows that KS distributional consistency is maintained across the full asset universe when paths are generated via the factor structure $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{SPY,t} + \hat{\eta}_{i,t}$. Panel (c) illustrates representative marginal density comparisons for three assets spanning a wide range of systematic risk exposure.

Table 1: Descriptive statistics and stylized-facts tests for SPY daily excess growth rates. In-sample: 2014–2024 ($T = 2,766$); out-of-sample: 2025 ($T = 249$). Gaussian and Laplace columns show MLE fits to the in-sample data. JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	—	—
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	—	—
LB test on $ G_t $ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	—	—

Table 2: In-sample and out-of-sample model comparison for SPY ($N = 100$, 1,000 simulated paths, significance level $\alpha = 0.05$). GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. HMM-NJ: standard Hidden Markov Model without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Kurtosis: mean excess kurtosis across simulated paths. Standard errors in parentheses: binomial SE for pass rates, $\text{std}/\sqrt{n}$ for kurtosis, bootstrap SE ($B = 500$) for ACF-MAE.

Metric	GARCH(1,1)	HMM-NJ ($N = 100$)	HMM-WJ ($N = 100$)
<i>In-sample: 2,766 trading days (2014–2024)</i>			
KS pass rate (%)	4.6 (0.7)	99.4 (0.2)	97.1 (0.5)
AD pass rate (%)	1.1 (0.3)	98.5 (0.4)	90.4 (0.9)
Excess kurtosis (observed)		7.715	
Excess kurtosis (simulated)	7.5 (0.32)	5.7 (0.02)	5.5 (0.03)
ACF-MAE (lags 1–252)	0.031	0.059	0.052
<i>Out-of-sample: 249 trading days (2025)</i>			
KS pass rate (%)	79.0 (1.3)	96.7 (0.6)	95.0 (0.7)
AD pass rate (%)	71.2 (1.4)	96.3 (0.6)	95.4 (0.7)
Excess kurtosis (observed)		6.867	
Excess kurtosis (simulated)	1.7 (0.07)	5.1 (0.07)	5.0 (0.08)
ACF-MAE (lags 1–252)	0.026	0.041	0.039
Parameters estimated	3	2	4

Table 3: Sensitivity of model performance to state resolution N . All metrics computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252. Standard errors in parentheses: binomial SE for pass rates, bootstrap SE ($B = 500$) for ACF-MAE.

Model	N	KS pass (%)	AD pass (%)	ACF-MAE
HMM-NJ	200	99.6 (0.2)	98.5 (0.4)	0.059
HMM-NJ	150	99.6 (0.2)	98.4 (0.4)	0.059
HMM-NJ	100	99.0 (0.3)	98.8 (0.3)	0.059
HMM-NJ	90	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	60	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	30	98.9 (0.3)	97.9 (0.5)	0.059
HMM-WJ	200	96.0 (0.6)	88.6 (1.0)	0.049
HMM-WJ	150	96.9 (0.5)	88.8 (1.0)	0.050
HMM-WJ	100	96.8 (0.6)	91.7 (0.9)	0.053
HMM-WJ	90	97.4 (0.5)	91.0 (0.9)	0.052
HMM-WJ	60	96.8 (0.6)	92.4 (0.8)	0.054
HMM-WJ	30	98.0 (0.4)	96.8 (0.6)	0.057

Table 4: Cross-sectional summary statistics for the Single-Index Model (SIM) extension across S&P 500 constituents. Individual asset paths are generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$ where $G_{\text{SPY},t}$ is drawn from HMM-WJ ($N = 100$) simulated paths and $\hat{\eta}_{i,t}$ is resampled from empirical residuals. KS pass rates computed over 1,000 paths at $\alpha = 0.05$.

Statistic	Mean	Median	5th pct	95th pct
<i>In-sample (424 assets, 2,766 trading days, 2014–2024)</i>				
KS pass rate (%)	58.4	66.7	2.5	96.0
SIM R^2	0.298	0.295	0.107	0.507
$\hat{\beta}$ (factor loading)	1.066	1.086	0.480	1.657
$\hat{\alpha}$ (intercept)	−0.0328	−0.0196	−0.2037	0.0964
<i>Out-of-sample (417 assets, 249 trading days, 2025)</i>				
KS pass rate (%)	82.1	91.8	23.8	99.0
SIM R^2	0.142	0.142	−0.169	0.479
$\hat{\beta}$ (factor loading)	1.064	1.084	0.482	1.660
$\hat{\alpha}$ (intercept)	−0.0320	−0.0186	−0.2035	0.0965

Algorithms

Algorithm 1 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f , Number of states N

Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\mathcal{Q}$

- 1: Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t .
 - 2: Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
 - 3: Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
 - 4: Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
 - 5: Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathcal{Q} = \{Q_k\}$.
 - 6: Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros.
 - 7: **for** $t = 1$ to $T - 1$ **do**
 - 8: Identify transition $i = s_t$ to $j = s_{t+1}$.
 - 9: $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
 - 10: **end for**
 - 11: **for** $i = 1$ to N **do**
 - 12: $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ {Row-wise normalization}
 - 13: **end for**
 - 14: **return** $\mathbf{T}, \mathcal{Q}$
-

Algorithm 2 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M

Ensure: Simulated state sequence $S_{1:M}$

```
1: Define tail-states  $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$  and  $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$ .
2: Sample initial state  $S_1 \sim \text{Categorical}(\bar{\pi})$ .
3: Set  $counter \leftarrow 2$ .
4: while  $counter \leq M$  do
5:   Sample  $u \sim \text{Uniform}(0, 1)$ .
6:   if  $u < \epsilon$  then
7:     Sample jump-duration  $K \sim \text{Poisson}(\lambda)$ .
8:     for  $j = 1$  to  $K$  while  $counter \leq M$  do
9:       Sample  $w \sim \text{Uniform}(0, 1)$ .
10:      if  $w < p_{neg}$  then
11:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$  {Bias toward negative-tail events}
12:      else
13:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ 
14:      end if
15:       $counter \leftarrow counter + 1$ .
16:    end for
17:  else
18:    Sample  $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1}, :})$ .
19:     $counter \leftarrow counter + 1$ .
20:  end if
21: end while
22: return  $S_{1:M}$ 
```

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Quantile boundaries $\mathcal{Q}$

Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

- 1: Define $z_{0.999} = 3.09$.
 - 2: **for** $k = 1$ to N **do**
 - 3: Set $\mu_k = (Q_{k-1} + Q_k)/2$.
 - 4: Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$.
 - 5: **end for**
 - 6: Initialize the reconstructed excess growth rate vector $\hat{\mathbf{R}}$ of length M .
 - 7: **for** $t = 1$ to M **do**
 - 8: Retrieve the simulated state $k = S_t$.
 - 9: Sample the continuous excess growth rate $\hat{R}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$.
 - 10: **end for**
 - 11: **return** $\hat{R}_{1:M}$
-

Supplemental Information

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$

Require: Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$

Require: Kurtosis penalty weight $w_K = 0.20$

Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

- 1: Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$.
 - 2: Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$.
 - 3: Initialize minimum error $E_{min} \leftarrow \infty$.
 - 4: **for** each ϵ in grid space **do**
 - 5: **for** each λ in grid space **do**
 - 6: Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$.
 - 7: **for** $p = 1$ to P **do**
 - 8: Simulate path using Algorithm 2 to obtain state sequence $S_{1:M}$.
 - 9: Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$.
 - 10: $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ .
 - 11: $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$.
 - 12: **end for**
 - 13: Compute ensemble averages: $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\bar{K}_{sim} = K_{sum}/P$.
 - 14: Compute objective: $J = \sum_{\tau=1}^L (A_{obs}(\tau) - \bar{A}_{sim}(\tau))^2 + w_K (K_{obs} - \bar{K}_{sim})^2$.
 - 15: **if** $J < E_{min}$ **then**
 - 16: $E_{min} \leftarrow J$
 - 17: $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$
 - 18: **end if**
 - 19: **end for**
 - 20: **end for**
 - 21: **return** ϵ^*, λ^*
-